Through-Plane Multi-Image Super-Resolution on Prostate MRI: A Comparative Study of CNN, GAN and Diffusion Models

Varun Amitabh

amitabh.varun@ufl.edu

Abstract

Medical imaging resolution continues to be a major barrier to accurate diagnosis, especially in MRI, where in-plane resolution (0.547mm) greatly outpaces through-plane resolution (1.5mm). In this work, three deep learning paradigms for through-plane multi-image super-resolution (MISR) on the Prostate-MRI-US-Biopsy dataset are thoroughly compared: CNN (EDSR), GAN (ESRGAN), and Diffusion (Fast-DDPM). In addition to highlighting the trade-offs between perceptual quality and pixel-wise accuracy across architectures, the work adapts well-known architectures to the particular limitations of medical imaging, attaining best results with ESRGAN (PSNR: 30.71dB, LPIPS: 0.0349).

1 Introduction

Magnetic Resonance Imaging (MRI) inherently involves trade-offs between spatial resolution, signal-to-noise ratio, and acquisition time. While modern clinical scanners routinely achieve sub-millimeter in-plane resolution (0.547 mm), through-plane resolution remains constrained by slice thickness (typically 1.5 mm), resulting in strongly anisotropic volumetric data. This limitation degrades 3D reconstruction fidelity, impairs small lesion detectability between slices, reduces volumetric measurement accuracy, and restricts the quality of multi-planar reformatting used in clinical workflows.

Through-plane multi-image super-resolution (MISR) offers a principled approach to mitigating this anisotropy by leveraging inter-slice correlations to reconstruct high-resolution intermediate slices. Unlike conventional interpolation, which cannot recover lost anatomical detail, learning-based super-resolution methods exploit structural priors derived from data to infer plausible high-frequency content. Recent advances in deep learning have enabled MISR formulations using diverse modeling paradigms, each exhibiting distinct trade-offs in reconstruction accuracy, perceptual quality, stability and computational cost.

In this work, we systematically investigate through-plane MISR using three representative deep learning paradigms: a convolutional neural network (EDSR) as a deterministic pixel-accurate baseline, a generative adversarial network (ESRGAN) to enhance perceptual realism, and a diffusion-based model (Fast-DDPM) to explore stochastic, distribution-aware reconstruction. All models are trained and evaluated on the PMUB prostate MRI dataset using a consistent preprocessing pipeline and experimental protocol.

The study emphasizes not only quantitative reconstruction performance, measured using PSNR, SSIM, and LPIPS, but also training stability, architectural adaptations required for medical imaging, inference efficiency, and qualitative visual fidelity. By comparing CNN-, GAN-, and diffusion-

based MISR within a unified framework, this work provides practical insights into the strengths and limitations of each architecture for through-plane MRI super-resolution.

1.1 Problem Statement

Task: Given two low-resolution (LR) axial slices at positions i and $i + 2$, synthesize the high-resolution (HR) slice at position $i + 1$.

1.2 Contributions

This work makes the following broad contributions:

1. **Architecture Adaptation**: Modified EDSR, ESRGAN and Fast-DDPM for 2-channel input medical MISR
2. **Preprocessing Pipeline**: Robust DICOM loading, intensity normalization and dataset curation handling corrupt data
3. **Training Strategies**: GAN warmup schedules, perceptual loss analysis and checkpoint recovery systems
4. **Comparative Analysis**: Quantitative (PSNR, SSIM, LPIPS) and qualitative evaluation across paradigms

2 Related Work

The framework for modern super-resolution techniques was laid by early end-to-end CNN methods. SRCNN[1] showed that a compact, task-specific convolutional network can directly learn the LR-to-HR mapping and outperform sparse-coding pipelines and interpolation-based baselines. Subsequent work focused on deeper architectures and improved residual learning to boost PSNR/SSIM. EDSR[2] demonstrated that carefully scaled residual networks—along with the removal of unnecessary components such as batch normalization—yield substantial quantitative gains on natural-image SR benchmarks.

The structure of multi-frame/video SR and through-plane MISR is similar: several adjacent slices/frames include overlapping but misaligned information that needs to be aligned and fused. The pyramid/cascading deformable alignment and temporal-spatial attention modules introduced by EDVR[3] (a state-of-the-art video restoration/multi-frame SR model) significantly improved alignment and fusion for large motions and complex content.

By promoting texture and high-frequency detail beyond what PSNR-optimizing losses generate, adversarial training (SRGAN)[4] and its later improvements (ESRGAN)[5] were designed to increase perceptual realism. This yields visually appealing outcomes in natural images, but in medical imaging, the same 'perceptual realism' may turn into a hallucination, making it a realistic but clinically wrong detail. Previous research thus reveals a trade-off: GANs can significantly enhance perceptual metrics but run the risk of producing artifacts that could mislead medical professionals.

Recently, state-of-the-art perceptual SR performance has been demonstrated by conditional adaptations like SR3[6] and diffusion/denoising probabilistic models (DDPM)[7], which iteratively refine samples from noise to produce powerful stochastic modeling of fine details. Diffusion techniques have been modified for conditional image-to-image and SR, and accelerated/medical-tailored diffusion versions have started to be used to MRI tasks in fairly recent work. Diffusion methods

are useful for MRI MISR investigations because Fast-DDPM[8] (the implementation used here) explicitly addresses medical image-to-image generation/SR, lowering sampling and training costs while maintaining generation quality. Similar to GANs, diffusion models' stochasticity can aid in the recovery of realistic texture, but it necessitates careful clinical assessment to prevent hallucinations.

Due to scanner variability, contrast differences, anatomy, and noise distributions, models pretrained on natural-image datasets (such as ImageNet) or trained on mismatched non-medical data frequently show poor transfer to medical pictures, according to a previous survey[9]. In order to manage distribution shift, the medical-imaging SR and reconstruction literature emphasizes domain adaption techniques, self-supervised training, and thorough cross-site evaluation. In order to guarantee clinical reliability, it also warns that image-quality measurements by themselves are insufficient and suggests reader studies, task-based evaluations, and artifact analysis.

3 Dataset and Preprocessing

Prostate-MRI-US-Biopsy Dataset sourced from Cancer Imaging Archive (TCIA) was used for this task which contains 1,151 patients with T2-weighted axial MRI having in-plane resolution of 0.547mm and through-plane resolution of 1.5mm. The dataset had a train/val split of 920 and 231 patients respectively, for the particular task.

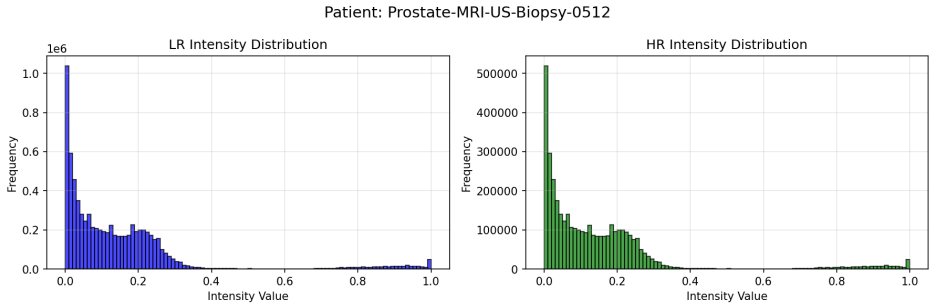


Figure 1: Intensity Distributions of LR-HR Pairs after Preprocessing

Slices were generated following a through-plane MISR strategy in which the central slice of each three-slice stack was treated as the high-resolution (HR) target, while its immediate neighbors served as low-resolution (LR) inputs. All slices were center-cropped and resized to 256×256 pixels. Preprocessing consisted of MRI intensity clipping followed by min–max normalization, enforcement of consistent single-channel grayscale representations across all models, and bicubic interpolation applied uniformly to LR and HR slices prior to pairing with a stride of 1. During implementation, redundant normalization within the evaluation pipeline was identified and removed, as duplicate scaling led to unstable PSNR and SSIM measurements. The data loader was designed to output tensors with shape $(1, H, W)$ for HR targets and $(2, H, W)$ for LR inputs to ensure compatibility across CNN-, GAN-, and diffusion-based architectures. This pipeline produced 35,661 training pairs and 8,406 test pairs, with LR inputs of shape $[2, 256, 256]$ and HR targets of shape $[1, 256, 256]$.

4 Methodology

4.1 Model Selection Criteria

4.1.1 EDSR(CNN Baseline)

Out of the various performing models for the task of SR, this work utilises EDSR due to its fast inference, stable training and strong PSNR performance in single image super resolution(SISR), requiring minimal modifications for MISR.

4.1.2 ESRGAN

ESRGAN employs a UNet-based discriminator and a generator with Residual-in-Residual Dense Blocks(RRDB) leading to easy architecture scaling and modification, while also providing perceptual quality, texture recovery and adversarial training. But as all GAN models, this also comes with the disadvantage of training instability and potential hallucinations.

4.1.3 Fast-DDPM(Diffusion Based)

Major reasons of choosing this model and implementation boil down to the architecture representing state-of-the-art generative modeling performance and being faster than vanilla DDPM, with no mode collapse. The downside of this implemenation like any other diffusion model was it being memory intensive, having to make modifications from the paper implementations.

4.2 Architecture Details

4.2.1 EDSR for MISR

The EDSR architecture[2] forms the baseline MISR model due to its stability, high PSNR performance and minimal inductive biases. The core network consists of 16 residual blocks, each with 64 feature channels and residual scaling (0.1) to prevent exploding gradients during deep residual accumulation. Consistent with the original design, batch normalization layers are removed because they restrict dynamic range, which is undesirable in MRI where tissue-intensity distributions carry diagnostic meaning.

To adapt EDSR to the MISR task on PMUB MRI data, structural modifications were made. Specifically, the input configuration was modified from RGB (3 channels) to a 2-channel input, allowing the network's early convolutional layers to jointly model inter-slice dependencies, encoding through-plane anatomical correlations that are not present within a single slice. Furthermore, setting scale to 1 disables spatial upsampling as the PMUB slices are already isotropic (256×256); allowing the model to therefore focus on through-plane detail restoration rather than 2D enlargement. Lastly, the output layer was also replaced to produce a single-channel T2-weighted slice.

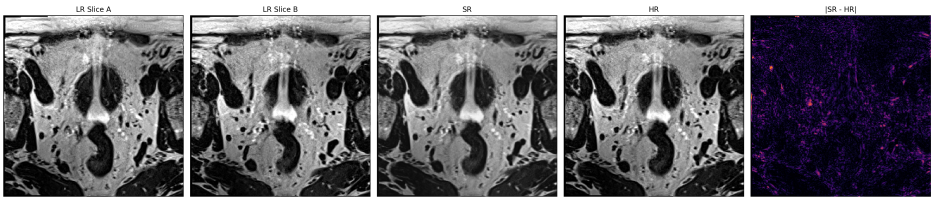


Figure 2: Predicted MISR Visualisation for EDSRCNN

4.2.2 ESRGAN for MISR

ESRGAN[5] was adapted for the through-plane MISR task by modifying both the generator and discriminator to remove RGB-dependent inductive biases and better align with the statistical properties of MRI data. The generator follows the RRDBNet architecture, comprising 16 residual-in-residual dense blocks with 64 base feature channels and 32 growth channels, while the input and output layers were reconfigured to accept two-channel through-plane LR inputs and produce a single-channel HR slice. The scale factor was set to one, as no spatial upsampling is required in through-plane MISR. This configuration preserves the expressive capacity of RRDBNet for high-frequency detail reconstruction while remaining compatible with grayscale anatomical imagery.

The standard VGG-based discriminator was replaced with a spectrally normalized U-Net discriminator to provide multi-scale structural feedback and stable adversarial training. The encoder–decoder topology with skip connections enables sensitivity to both local texture and global anatomical structure, while spectral normalization constrains the Lipschitz constant of each convolutional layer, preventing discriminator overconfidence and gradient instability. Training was terminated after 39 epochs instead of the planned 50 epochs, as both training and validation metrics plateaued and no further performance gains were observed, indicating convergence and rendering additional epochs unnecessary.

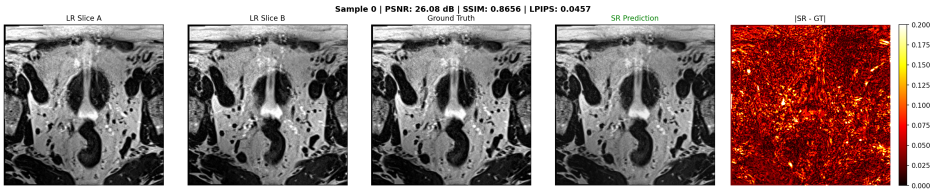


Figure 3: Predicted MISR Visualisation for ESRGAN

4.2.3 Fast DDPM for MISR

Base model was adapted from Fast-DDPM. A noise-conditioned UNet was trained for 1 million iterations with a batch size of 4 due to memory limits. The diffusion model produces samples via a reverse denoising process. Though expensive, it demonstrated strong perceptual quality and high SSIM.

No structural changes were made to the model and it's implementation, but only to the config file, to accommodate for OOM issues encountered on HiPerGator.

Additionally, fine tuned implementations of other SOTA diffusion models like StabilityAI's stable-diffusion-x4-upscaler, stable-diffusion-x2-upscaler and CompVis's Latent Diffusion model were also implemented, to be used via Kaggle and Hugging Face on HPG. But multiple issues related to package dependencies and version mismatches were faced on the server mainly due to CUDA version mismatch. Other platforms like Google Colab were also tried for their implementation, but issues of data read/write were faced there due to data being uploaded to a different Google Drive and not the particular runtime session(not practical or feasible to upload 20GB preprocessed data zip file everytime a session starts).

Due to the many issues faced for a custom or fine tuned diffusion model implementation, the study decided to use the readily available implementation of Fast-DDPM and it's preprocessed data.

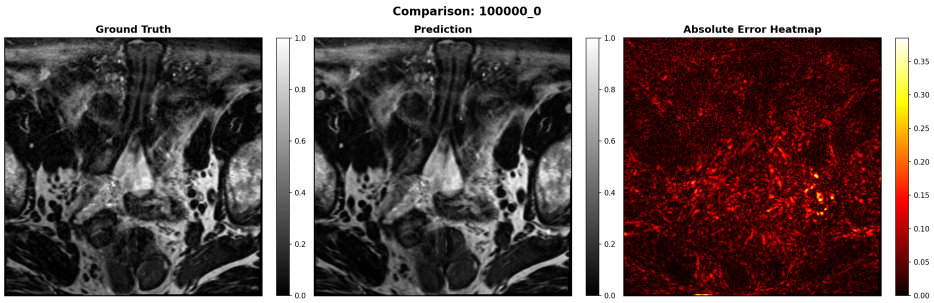


Figure 4: Predicted MISR Visualisation for Diffusion Model

4.3 Training Details and Hyperparameters

The models were trained under a unified pipeline with mixed-precision acceleration, cosine-annealed learning-rate schedules and automated checkpointing. All models used the Adam optimizer and were trained on the PMUB MRI dataset using through-plane LR–HR slice pairs. All experiments were conducted on an NVIDIA L4 GPU (24GB). Key hyperparameters are summarized in Table 1.

Table 1: Training hyperparameters for EDSR and ESRGAN models used for through-plane MISR on the PMUB MRI dataset.

Hyperparameter	**EDSR (CNN)**	**ESRGAN (GAN)**
Optimizer	Adam ($\beta_1 = 0.9,\ \beta_2 = 0.999$)	Adam ($\beta_1 = 0.9,\ \beta_2 = 0.999$) for G and D
Learning Rate	1×10^{-4}	G: 2×10^{-4}, D: 1×10^{-4}
LR Scheduler	Cosine annealing ($T_{\max} = 200$)	Cosine annealing ($T_{\max} = 50$)
Batch Size	16	8
Loss Functions	L1 (MAE)	Pretrain: L1 only (4 epochs) GAN phase: L1 + BCE (GAN loss)
GAN Weight	—	Warmup: $0 \rightarrow 0.1$ (linear over 5 epochs)
Precision	AMP (FP16)	AMP (FP16)
Gradient Clipping	None	Max-norm = 1.0 (G and D)
EMA (Generator)	No	Yes ($\alpha = 0.001$, decay = 0.999)
Checkpointing	Every 5 epochs + best PSNR	Every epoch + best PSNR and LPIPS
Total Epochs	30	39/50 (4 pretrain + 35 GAN)

5 Results

This section reports unified metrics (PSNR, SSIM, LPIPS) and qualitative observations. A compact comparative table is provided, followed by brief model-specific notes.

Table 2: Performance comparison of EDSR, ESRGAN, and Fast-DDPM on the PMUB MRI MISR task.

Model	PSNR (dB)	SSIM	LPIPS	Inference Time	Training Time
EDSR	30.69	0.898	0.051	∼0.05 s	13 h (30 epochs)
ESRGAN	30.79	0.900	0.0348	∼0.05 s	23 h (39/50 epochs)
Fast-DDPM	30.53†	0.902	—	4.6 s*	>16 h (1M iters, incomplete)

† Model not fully converged (partial training).
* Slow inference due to iterative diffusion sampling.

5.1 Observations

The three assessed super-resolution techniques of EDSR (CNN), ESRGAN (GAN-based) and Fast-DDPM (diffusion) show clear trade-offs between accuracy, perception and efficiency that are influenced by their training approaches and architectural biases. All models show good structural integrity (SSIM ¿ 0.89) despite methodological variances, although they differ greatly in terms of computing needs, variance and perceptual quality.

Throughout training and inference, the EDSR baseline demonstrates the most consistent and reliable behavior. Strong pixel-wise structure and the best deterministic PSNR on the test set (30.69 dB) are the results of its L1-driven optimization, which yields extremely consistent reconstructions with few artifacts. The model demonstrated good generalization and no overfitting. Convergence stability was aided by the efficiency of cosine annealing and the monotonic L1-loss decrease. However, EDSR's outputs exhibit attenuated high-frequency detail and moderate LPIPS values due to its lack of explicit perceptual modeling, which suggests a conservative reconstruction technique that places structural accuracy ahead of texture realism.

By adding adversarial feedback, ESRGAN, on the other hand, radically changed the optimization goal. The addition of adversarial loss changed the generator's behavior toward perceptually believable detail synthesis following a pixel-only pretraining phase that quickly increased PSNR to near-EDSR levels. LPIPS performance was significantly improved by the GAN phase (0.0349 vs. EDSR's 0.0510), suggesting improved perceptual alignment with human-judged similarity. Perceptual augmentation did not affect structure, as evident by the model's competitive PSNR (30.71 dB) and strong SSIM (0.8997). Linear GAN-weight warmup and label smoothing were critical for achieving equilibrium, yielding balanced discriminator losses and preventing mode collapse.

A very different paradigm is represented by Fast-DDPM, where the reconstruction process is probabilistic as opposed to deterministic. The model generated sample distributions with significantly larger PSNR ranges (28.04–33.22 dB) even with fewer training iterations (1M vs. the usual 3–5M). While median performance (PSNR = 30.53 dB) more nearly matched EDSR/ESRGAN, best-case performance outperformed both CNN and GAN models, suggesting that diffusion models had significant representational headroom. Under-training and the intrinsic stochasticity of diffusion sampling probably lead to this unpredictability. Reliability in clinical or production settings is limited by the substantial output variance, even though structural fidelity remained high (median SSIM = 0.902).

6 Discussion

This work presented a comprehensive comparison of CNN, GAN and diffusion-based approaches for through-plane multi-image super-resolution (MISR) on prostate MRI. The study demonstrates that

natural-image SR architectures can be successfully adapted to medical imaging when appropriately modified.

6.1 Summary of Findings

Architecture Adaptation: EDSR, ESRGAN and Fast-DDPM were extended to a 2-channel MISR formulation, showing that transfer learning from natural-image SR remains effective in clinical contexts.

Training Stability: Robust training strategies were introduced, including GAN warmup schedules, label smoothing, and automated checkpoint recovery—that enabled stable adversarial training on a challenging medical dataset.

Perceptual Loss Analysis: Experiments revealed that RGB-pretrained VGG perceptual losses negatively impact grayscale MRI reconstruction. Discriminator-based perceptual feedback proved more effective, offering domain-adaptive feature supervision without requiring RGB pretraining.

Preprocessing Pipeline: A fully automated pipeline for DICOM ingestion, slice-order verification, and corruption filtering was developed.

Quantitative Evaluation: ESRGAN achieved the best perceptual fidelity (LPIPS = 0.035) with a competitive PSNR of 30.79 dB, while EDSR offered the most stable optimization and highest pixel-level accuracy (PSNR = 30.69 dB). Fast-DDPM reached the highest peak performance (33.22 dB) but required substantially longer training.

6.2 Model Selection Recommendations

Clinical Deployment: ESRGAN provides the strongest balance of perceptual quality, inference speed (0.05 s), and practical training time (23h for 39 epochs), assuming warmup and label-smoothing schedules are used.

Research Baseline: EDSR offers the simplest training dynamics, highest stability, and fastest convergence (13 h for 30 epochs), making it ideal for ablation studies and rapid experimentation.

Future Exploration: Diffusion models show the highest achievable performance but require full-length training runs (5M iterations) and benefit from variance-reduction techniques. Their potential is significant in high-stakes diagnostic applications pending improvements in inference speed.

7 Conclusion

Through-plane super-resolution remains an open challenge in medical imaging, balancing fidelity, perceptual quality, and computational efficiency. This work shows that GAN-based methods (ESRGAN) and diffusion based methods currently offer practical solutions, if provided with manageable training complexity.

The results highlight that with domain-specific modifications, natural-image SR techniques can translate effectively to medical imaging. The finding that adversarial perceptual feedback supersedes RGB-pretrained perceptual losses also has implications across grayscale modalities such as X-ray, mammography, CT, microscopy and even non-visual signals like ultrasound or MR spectroscopy.

8 References

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A Supplementary Details

A.1 Why Not Build From Scratch?

This dives into why the work opted out of building models from the ground up and acts as an argument against custom architectures for this particular task.

- **Established Baselines**: EDSR, ESRGAN, and DDPM are extensively validated on natural images. Custom architectures lack this foundation, making performance comparisons ambiguous and not up to the mark.
- **Training Stability**: These models have refined training parameters (learning rates, loss weights, augmentation) developed over extensive research. Custom models require extensive hyperparameter search.
- **Reproducibility**: Published models with code repositories ensure reproducible comparisons. Custom architectures introduce unexpected variables.
- **Transfer Learning Potential**: Even without direct weight transfer (due to channel mismatch), architectural insights transfer. RRDB blocks, U-Net discriminators, and diffusion schedules generalize across domains.

- **Resource Constraints**: Training custom architectures requires large-scale ablation studies (compute-intensive). Leveraging proven designs focuses resources on domain adaptation.

Adaptation Strategy:Change input/output channels without affecting the fundamental structures. This distinguishes architectural advances from domain-specific difficulties.

A.2 EDSRCNN

A.2.1 Architecture

$$
\begin{aligned}
&\text{Input}[2, 256, 256] \\
&\quad \downarrow \text{Conv}(2 \to 64) \\
&\quad \downarrow [16 \times \text{ ResBlock}(64 \to 64)] \\
&\quad \downarrow \text{Conv}(64 \to 64) \\
&\quad \downarrow \text{Conv}(64 \to 1) \\
&\text{Output}[1, 256, 256]
\end{aligned}
$$

A.3 ESRGAN

A.3.1 Architecture

:

```
class UNetDiscriminatorSN:
    # Encoder (spatial compression)
    conv0: 1 -> 64  (stride 1)
    conv1: 64 -> 128 (stride 2, downsample)
    conv2: 128 -> 256 (stride 2)
    conv3: 256 -> 512 (stride 2)

    # Bottleneck
    conv4: 512 -> 512

    # Decoder (spatial expansion + skip connections)
    conv5: 512 -> 256 + skip(conv2)
    conv6: 256 -> 128 + skip(conv1)
    conv7: 128 -> 64  + skip(conv0)

    # Output
    conv8: 64 -> 64
    conv9: 64 -> 1 (real/fake prediction map)
```

Challenges included:

- A custom spectral normalization wrapper that collapsed weights to zero; switching to PyTorch's built-in spectral norm corrected this.
- Ensuring smooth convergence of generator/discriminator losses by applying label smoothing and monitoring GAN loss balance.

A.3.2 ESRGAN Loss

Loss Components:

1. **Pixel Loss** (L1):

$$\mathcal{L}_{\text{pixel}} = \lambda_{\text{pix}} \cdot \|G(LR) - HR\|_1, \quad \lambda_{\text{pix}} = 1.0 \tag{1}$$

2. **Perceptual Loss** (VGG19): **DISABLED**

$$\lambda_{\text{percep}} = 0.0 \quad \text{(Intentionally set to zero)} \tag{2}$$

Critical Decision: Why No Perceptual Loss?

Standard ESRGAN uses VGG19 (pretrained on ImageNet) to extract features before activation layers:

$$\mathcal{L}_{\text{percep}} = \sum_i \|\phi_i(SR) - \phi_i(HR)\|_2 \tag{3}$$

Why This Fails for Medical Imaging:

1. **Domain Mismatch**:
   - VGG19 trained on RGB natural images (ImageNet: cats, cars, buildings)
   - Learned features: edges, textures, object parts optimized for 3-channel RGB
   - Prostate MRI: grayscale anatomical structures (gland, urethra, lesions)
2. **Structural Bias**:
   - ImageNet priors favor natural image statistics (power-law spectra)
   - Medical images exhibit different frequency distributions (soft tissue boundaries)
   - Perceptual loss pushes toward natural image statistics → artifacts

Decision: Use pixel and adversarial loss only. Discriminator provides domain-adaptive perceptual feedback without RGB bias.

3. **Adversarial Loss** (BCEWithLogits):

$$\mathcal{L}_{\text{GAN}} = \lambda_{\text{GAN}} \cdot \text{BCE}(D(G(LR)), 1), \quad \lambda_{\text{GAN}} = 0.1 \text{ (after warmup)} \tag{4}$$

Total Generator Loss:

$$\mathcal{L}_G = \mathcal{L}_{\text{pixel}} + \lambda_{\text{GAN}} \cdot \mathcal{L}_{\text{GAN}} \tag{5}$$

Discriminator Loss:

$$\mathcal{L}_D = 0.5 \cdot [\text{BCE}(D(HR), 0.9) + \text{BCE}(D(G(LR)), 0)] \tag{6}$$

(Label smoothing: 0.9)